Mathematics: analysis and approaches
Standard level
Paper 1
4.1–4.6 Statistics and probability

13 October 2026

W4ss Academy practice set 1 | **Topic** 4.1–4.6 Statistics and probability | **Ref.** W4SS–MAA–T4–P1–01

Student name

30 minutes

Instructions to students

- Write your name in the boxes above.
- Do not open this examination paper until instructed to do so.
- You are **not** permitted access to any calculator for this paper.
- Section A: answer all questions. Answers must be written within the answer boxes provided.
- Section B: answer all questions on the answer pages at the end of this paper.
- Unless otherwise stated in the question, all numerical answers should be given exactly or correct to three significant figures.
- A clean copy of the **mathematics: analysis and approaches formula booklet** is required for this paper.
- The maximum mark for this examination paper is **[25 marks]**.

Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

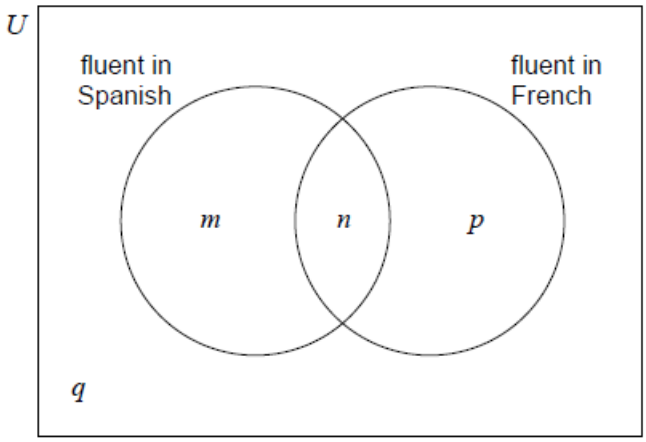
Section A

Answer **all** questions. Answers must be written within the answer boxes provided. Working may be continued below the lines, if necessary.

1. [Maximum mark: 6]

In a class of 30 students, 18 are fluent in Spanish, 10 are fluent in French, and 5 are not fluent in either of these languages. The following Venn diagram shows the events “fluent in Spanish” and “fluent in French”.

The values m, n, p and q represent numbers of students.



(a) Write down the value of q. [1]

(b) Find the value of n. [2]

(c) Write down the value of m and of p. [3]

(This question continues on the following page)

(Question 1 continued)

2. [Maximum mark: 5]

Box 1 contains 5 red balls and 2 white balls.

Box 2 contains 4 red balls and 3 white balls.

(a) A box is chosen at random and a ball is drawn. Find the probability that the ball is red. [3]

(b) Let A be the event that "box 1 is chosen" and let R be the event that "a red ball is drawn".

Determine whether events A and R are independent. [2]

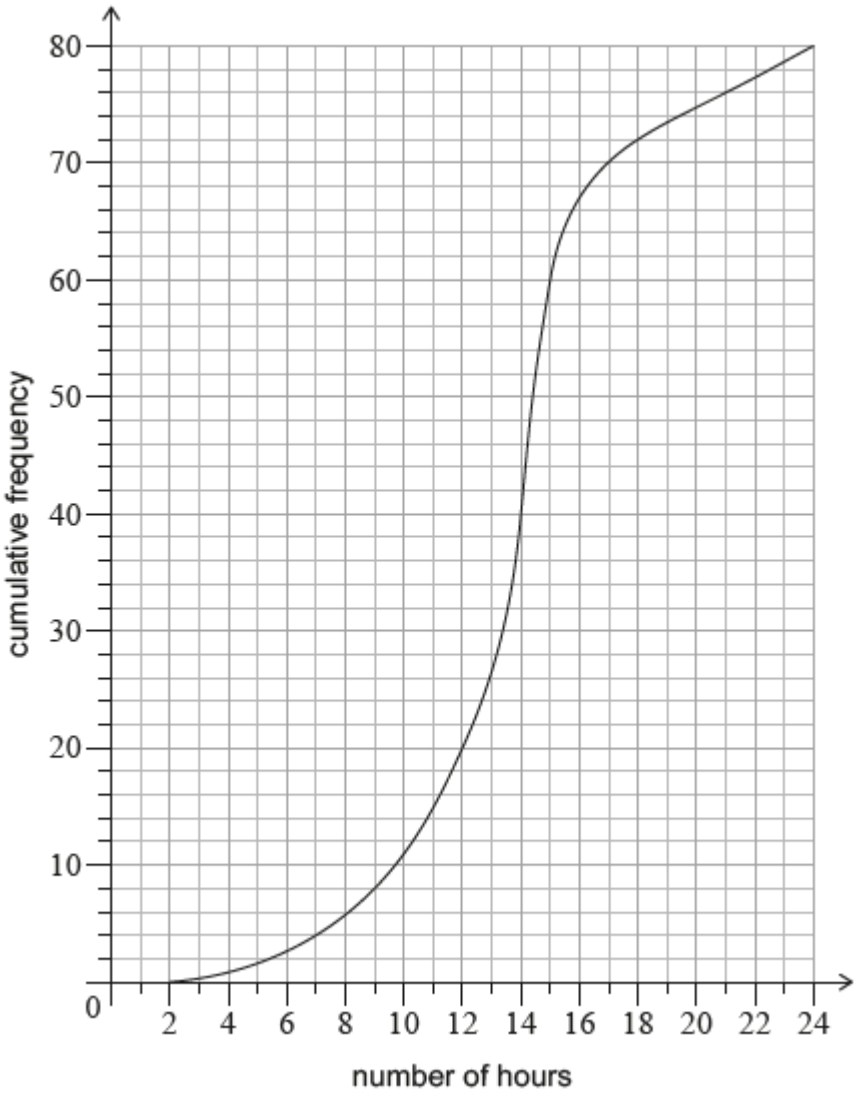
Section B

Answer **all** questions on the answer pages at the end of this paper. Please start each question on a new page.

3. [Maximum mark: 14]

A large school has students from Year 6 to Year 12.

A group of 80 students in Year 12 were randomly selected and surveyed to find out how many hours per week they each spend doing homework. Their results are represented by the following cumulative frequency graph.



(a) Find the median number of hours per week these Year 12 students spend doing homework. [2]

(b) Given that 10% of these Year 12 students spend more than k hours per week doing homework, find the value of k. [3]

(This question continues on the following page)

Do not write solutions on this page.

(Question 3 continued)

This same information is represented by the following table.

Hours (h) spent doing homework	$2 < h \leq 7$	$7 < h \leq 15$	$15 < h \leq 21$	$21 < h \leq 24$
Frequency	4	p	16	q

(c) Find the value of p and the value of q. [4]

There are 320 students in Year 12 at this school.

(d) Estimate the number of Year 12 students that spend more than 15 hours each week doing homework. [3]

(e) (i) Explain why this sampling method might not provide an accurate representation of the amount of time **all** of the students in the school spend doing homework. [1]

(ii) Suggest a more appropriate sampling method. [1]

Do not write solutions on this page.

Section B — answers to question 3

Section B — answers to question 3 (continued)

Sources of the questions

This practice paper was put together by W4ss Academy for internal class use. The questions are taken from the IB DP Mathematics: analysis and approaches Questionbank (standard level, topic 4 Statistics and probability) and cover subtopics 4.1 to 4.6. They remain the property of the International Baccalaureate Organization and must not be published or sold.

Section A — question 1	19N.1.SL.TZ0.S_2 (former Mathematics SL course) · 4.6 Venn diagrams
Section A — question 2	21N.1.SL.TZ0.4 · 4.5, 4.6 tree diagrams, independent events
Section B — question 3	21M.1.SL.TZ2.7 · 4.2 cumulative frequency, percentiles · 4.3 median · 4.1 sampling